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United States Patent Application Publication

20250283691

Kind Code

A1

Publication Date

September 11, 2025

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HANDGUN BRACE

Abstract

A brace for stabilizing a handheld device such as a handgun against a user's forearm includes a vertically elongated body adapted to detachably engage a portion of the handheld device, and an arm pivotably connected to a portion of the body. The pivotable arm is selectably moveable between a stowed position wherein the arm is vertically aligned with the body, and a deployed position wherein the arm provides a surface against which a user's forearm is removably receivable to stabilize the handheld device when the body is engaged with the handheld device.

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Family ID: 72041879

Appl. No.: 19/211127

Filed: May 16, 2025

Related U.S. Application Data

parent US continuation 18228295 20230731 parent-grant-document US 12332017 child US 19211127

parent US continuation 17035210 20200928 parent-grant-document US 11754367 child US 18228295

parent US continuation 16701519 20191203 parent-grant-document US 10823528 child US 17035210

parent US continuation 15485345 20170412 parent-grant-document US 10180302 child US 16211522

parent US continuation 15407362 20170117 parent-grant-document US 9664477 child US 15485345

parent US continuation-in-part 16211522 20181206 parent-grant-document US 10690442 child US 16701519

Publication Classification

Int. Cl.: F41C27/22 (20060101); F41A23/02 (20060101); F41C23/04 (20060101); F41C23/12 (20060101)

U.S. Cl.:

CPC F41C27/22 (20130101); F41A23/02 (20130101); F41C23/04 (20130101); F41C23/12 (20130101);

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This application is a continuation of co-pending U.S. patent application Ser. No. 18/228,295, filed Jul. 31, 2023 and entitled “HANDGUN BRACE,” which is a continuation of U.S. patent application Ser. No. 17/035,210, filed Sep. 28, 2020 and entitled “HANDGUN BRACE,” which issued as U.S. Pat. No. 11,754,367 on Sep. 12, 2023 and which is a continuation application of U.S. patent application Ser. No. 16/701,519, filed Dec. 3, 2019 and entitled “HANDGUN BRACE,” which issued as U.S. Pat. No. 10,823,528 on Nov. 3, 2020 and which is a continuation-in-part application of U.S. patent application Ser. No. 16/211,522, filed on Dec. 6, 2018 and entitled “HANDGUN BRACE,” which issued as U.S. Pat. No. 10,690,442 on Jun. 23, 2020 and which is a continuation application of U.S. patent application Ser. No. 15/485,345, filed on Apr. 12, 2017 and entitled “HANDGUN BRACE,” which issued as U.S. Pat. No. 10,180,302 on Jan. 15, 2019 and which is a continuation application of U.S. patent application Ser. No. 15/407,362, filed on Jan. 17, 2017 entitled “HANDGUN BRACE,” which issued as U.S. Pat. No. 9,664,477 on May 30, 2017 and which claims priority to U.S. Provisional Patent Application Ser. No. 62/279,201 entitled “ARM BRACE DEVICE FOR A FIREARM,” filed on Jan. 15, 2016, and U.S. Provisional Patent Application Ser. No. 62/327,219 entitled “ARM BRACE AND LATCH ASSEMBLY FOR A FIREARM,” filed on Apr. 25, 2016, all of which are hereby incorporated by reference in their entirety.

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STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0003] Not Applicable

REFERENCE TO SEQUENCE LISTING OR COMPUTER PROGRAM LISTING APPENDIX

[0004] Not Applicable

BACKGROUND OF THE INVENTION

[0005] The present disclosure relates generally to a stabilizing device for a handgun or other handheld device and, more particularly, to an arm brace that receives and engages the user's forearm to stabilize the handgun or other handheld device during use.

[0006] The accuracy, speed, and precision with which a handgun can be reliably fired depend greatly upon the user's ability to hold the handgun steady while aiming and discharging it. This can be particularly challenging for physically disabled persons who cannot firmly grip or steady a handgun for a period of time sufficient to accurately fire the handgun. The difficulty of holding a

handgun in a steady position is heightened during one-handed operation because a user must bear much of the weight and recoil of the handgun with the wrist of one hand, which can cause the user's hand and wrist to tire prematurely and negatively impact the user's ability to safely and reliably handle and fire the handgun. This problem is further increased with large frame handguns in which the weight of the handgun is centered at a location forward of the grip because such handguns require the user to continuously exert a counterbalancing force on the grip to maintain the handgun in a steady firing position and prevent the muzzle of the handgun from tilting downward.

[0007] A number of devices designed to attach to a handgun to aid a user in holding or stabilizing the handgun are known. For example, U.S. Pat. No. 8,869,444 discloses an flexible cuff that attaches to a handgun and employs a pair of elongated spaced flaps between which a user's forearm is secured with an adjustable securement strap to help stabilize the handgun during firing.

However, the flaps apply oppositely directed inward forces to grip a user's forearm, which can chafe the user's forearm during prolonged use, while the strap requires the use of a second hand to operate, which introduces an undesirable degree of complexity that may be difficult for physically disabled users to overcome.

[0008] By contrast, the device marketed at the time of filing as the SHOCKWAVE® Blade Pistol Stabilizer employs a vertical stabilizing fin that rests against the inside of a user's forearm to stabilize an attached handgun during firing. Although designed with apertures through which a standard rifle sling may optionally be threaded to secure the fin to the user's forearm, in the absence of an accompanying sling or securement strap, the device relies solely on friction between the user's forearm and the fin to stabilize the handgun vertically during firing, which may be insufficient to adequately stabilize large frame handguns.

[0009] Additional handheld devices and apparatuses which tend to be similarly difficult and cumbersome to operate include metal detectors, vacuums, drills and various tools, among others. Such devices are often so bulky and heavy that normal operation can prematurely strain and tire the hand and wrist of even able-bodied persons. This is particularly disadvantageous for persons employed in positions where prolonged operation of the handheld device is necessary or unavoidable because premature strain can negatively impact the user's ability to continually and precisely operate the device.

[0010] Accordingly, what is need are improvements in stabilizing devices for handguns and other handheld devices.

SUMMARY OF THE INVENTION

[0011] This Brief Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0012] The presently disclosed subject matter overcomes some or all of the above-identified deficiencies of the prior art, as will become evident to those of ordinary skill in the art after a study of the information provided in this document. custom-character

[0013] Accordingly, in one aspect, the present disclosure provides a handgun brace for stabilizing a handgun on a user's forearm, the handgun brace including a body adapted to detachably engage a support structure of a handgun, such as an integral receiver extension. The handgun brace further includes a first arm extending from the body, and a second arm pivotally connected to the first arm such that the second arm is moveable between an open position and a closed position. When the second arm is in the closed position, a portion of the second arm is received against a portion of the body opposite the first arm. When the second arm is in the open position, the first arm and the second arm form a support surface shaped to receive and engage a user's forearm such that the user's forearm is removably receivable against the support surface to stabilize the handgun when the body is engaged with the support structure of the handgun.

[0014] In another aspect, the disclosure provides a telescopic handgun brace for stabilizing a handgun having a support structure extending rearwardly from a rear end thereof on a user's forearm. The handgun brace includes a mounting body having a front end, a rear end, a main passage extending axially within said body through at least the front end thereof, and at least one secondary passage extending axially through said body from said front end to said rear end, the support structure of said handgun being receivable by said main passage, and a main body having a front end, a rear end, and at least one rod extending forwardly from the front end of the main body, said at least one rod being telescopically receivable by said at least one secondary passage of the mounting body. The handgun brace can further have a first arm extending downwardly from said main body, and a second arm pivotally connected to the first arm. The second arm is selectively moveable between a stowed position wherein a portion of the second arm is received against the main body, and a deployed position wherein said first arm and said second arm form a support surface upon which a user's forearm is received during discharge of the handgun while said handgun brace is attached thereto.

[0015] In yet another aspect, the disclosure provides a telescopic handgun brace for stabilizing a handgun on a user's forearm, the handgun brace including a support structure having a forward end adapted to be detachably engaged with a rear end of a handgun, a rearward end opposite the forward end, a length, and a downwardly protruding rib extending longitudinally along a portion of the length, the rib having a plurality of cross notches formed in a side thereof. The handgun brace also includes a body having a front end, a rear end, a passage extending axially within the body through at least the front end thereof, said support structure being telescopically receivable by said passage; a forearm support member in which a user's forearm is removably receivable to stabilize the handgun when the rearward end of the support structure is received in said passage and the forward end is engaged with the rear end of said handgun, said forearm support member extending from said body; and a push-button locking latch extending laterally through a portion of the body, said latch adapted to releasably engage at least one of said cross notches to selectively lock said body in one of a plurality of positions relative to the support structure.

[0016] In another aspect, the disclosure provides a brace for stabilizing a handheld device against a user's forearm, the brace including a vertically elongated body adapted to detachably engage a portion of the handheld device, and an arm pivotably connected to the elongated body, the arm selectably moveable between a stowed position wherein the arm is substantially aligned with the elongated body, and a deployed position wherein the arm provides a surface against which a user's forearm is removably receivable to stabilize the handheld device when the elongated body is engaged with the handheld device.

[0017] Numerous other objects, advantages and features of the present disclosure will be readily apparent to those of skill in the art upon a review of the following drawings and description of a preferred embodiment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] Non-limiting and non-exhaustive embodiments are described with reference to the following figures.

[0019] FIG. 1 is a side elevational view of an embodiment of a brace for stabilizing a handgun or other handheld device constructed in accordance with the present disclosure showing the brace in use and attached to a handgun.

[0020] FIG. 2 is perspective view of the brace of FIG. 1.

[0021] FIG. 3 is a rear elevational view of the brace of FIG. 1 illustrating the brace's engagement with a user's forearm.

[0022] FIG. 4 is a front elevational view of the brace of FIG. 1.
[0023] FIG. 5A is a left side elevational view of the brace of FIG. 1.
[0024] FIG. 5B is a perspective view of the brace of FIG. 5A.
[0025] FIG. 6A is a right side elevational view of the brace of FIG. 1.
[0026] FIG. 6B is a perspective view of the brace of FIG. 6A.
[0027] FIG. 7A is a top plan view of the brace of FIG. 1.
[0028] FIG. 7B is a perspective view of the brace of FIG. 7A.
[0029] FIG. 8A is a bottom plan view of the brace of FIG. 1.
[0030] FIG. 8B is a perspective view of the brace of FIG. 8A.
[0031] FIG. 9 is a perspective view of the brace of FIG. 1 in a partially closed configuration showing the second arm positioned to be releasably engaged with the body of the brace.
[0032] FIG. 10 is a partially exploded perspective view of the brace of FIG. 1 in a non-use configuration.
[0033] FIG. 11 is a perspective view of a brace constructed in accordance with an alternate embodiment of the present invention showing the brace attached to a handgun and in a compact, non-use configuration.
[0034] FIG. 12 is a perspective view of the brace of FIG. 11 showing the brace in an extended, deployed configuration.
[0035] FIG. 13 is a partially exploded front perspective view of the brace of FIG. 11.
[0036] FIG. 14 is a right side perspective view of a brace constructed in accordance with another alternate embodiment of the present invention showing the brace in a non-use configuration.
[0037] FIG. 15 is a right side perspective view of the brace of FIG. 14 showing the brace in a deployed configuration.
[0038] FIG. 16 is an exploded view of the brace of FIG. 14.
[0039] FIG. 17A is a right side perspective view of the brace of FIG. 14.
[0040] FIG. 17B is another right side perspective view of the brace of FIG. 14.
[0041] FIG. 18 is a left side perspective view of the brace of FIG. 14.
[0042] FIG. 19 is a right side elevational view of the brace of FIG. 14.
[0043] FIG. 20 is a left side elevational view of the brace of FIG. 14.
[0044] FIG. 21 is a front elevational view of the brace of FIG. 14.
[0045] FIG. 22 is a rear elevational view of the brace of FIG. 14.
[0046] FIG. 23 is a bottom plan view of the brace of FIG. 14.
[0047] FIG. 24 is a top plan view of the brace of FIG. 14.
[0048] FIG. 25 is a bottom perspective view of an embodiment of a support structure for the brace of FIG. 14.
[0049] FIG. 26 is a perspective view of an embodiment of a latch for the brace of FIG. 14.
[0050] FIG. 27 is a sectional view of the brace of FIG. 14 showing the push button arm release in an unlocked position and the latch in a locked position.
[0051] FIG. 28 is a sectional view of the brace of FIG. 14 showing the push button arm release in a locked position and the latch in an adjustment position.
[0052] FIG. 29 is a sectional view of the brace of FIG. 14 showing the push button arm release in a locked position and the latch in a removal position.
[0053] FIG. 30 is another left side perspective view of the brace of FIG. 14.
[0054] FIG. 31 is a rear right side perspective view of the brace of FIG. 15.
[0055] FIG. 32 is a rear left side perspective view of the brace of FIG. 15.
[0056] FIG. 33 is a front right side perspective view of the brace of FIG. 15.
[0057] FIG. 34 is a bottom right side perspective view of the brace of FIG. 15.
[0058] FIG. 35 is a bottom left side perspective view of the brace of FIG. 15.
[0059] FIG. 36 is a right side elevational view of the brace of FIG. 15.
[0060] FIG. 37 is a left side elevational view of the brace of FIG. 15.

[0061] FIG. **38** is a front elevational view of the brace of FIG. **15**.
[0062] FIG. **39** is a rear elevational view of the brace of FIG. **15**.
[0063] FIG. **40** is a bottom plan view of the brace of FIG. **15**.
[0064] FIG. **41** is a top plan view of the brace of FIG. **15**.
[0065] FIG. **42** is an elevated rear right side perspective view of a brace for stabilizing a handheld device constructed in accordance with another alternate embodiment of the present invention showing the brace in a deployed configuration for a right-handed user.
[0066] FIG. **43** is a depressed front right side perspective view of the brace of FIG. **42**.
[0067] FIG. **44** is an elevated rear left side perspective view of the brace of FIG. **42**.
[0068] FIG. **45** is an elevated front left side perspective view of the brace of FIG. **42**.
[0069] FIG. **46** is another right side perspective view of the brace of FIG. **42**.
[0070] FIG. **47** is a right side elevational view of the brace of FIG. **42**.
[0071] FIG. **48** is a left side elevational view of the brace of FIG. **42**.
[0072] FIG. **49** is a front elevation view of the brace of FIG. **42**.
[0073] FIG. **50** is a rear elevational view of the brace of FIG. **42**.
[0074] FIG. **51** is a bottom plan view of the brace of FIG. **42**.
[0075] FIG. **52** is a top plan view of the brace of FIG. **42**.
[0076] FIG. **53** is an elevated rear right side perspective view of the brace of FIG. **42** showing the brace in a stowed configuration.
[0077] FIG. **54** is a depressed front right side perspective view of the brace of FIG. **53**.
[0078] FIG. **55** is an elevated rear left side perspective view of the brace of FIG. **53**.
[0079] FIG. **56** is an elevated front left side perspective view of the brace of FIG. **53**.
[0080] FIG. **57** is a right side elevational view of the brace of FIG. **53**.
[0081] FIG. **58** is a left side elevational view of the brace of FIG. **53**.
[0082] FIG. **59** is a front elevational view of the brace of FIG. **53**.
[0083] FIG. **60** is a rear elevational view of the brace of FIG. **53**.
[0084] FIG. **61** is a bottom plan view of the brace of FIG. **53**.
[0085] FIG. **62** is a top plan view of the brace of FIG. **53**.
[0086] FIG. **63** is an exploded perspective view of the brace of FIG. **56**.
[0087] FIG. **64** is a cutaway view of the brace of FIG. **58** taken long line **64-64** showing the brace in an unlocked position.
[0088] FIG. **65** is a cutaway view of the brace of FIG. **58** taken long line **65-65** showing the brace in a locked position.
[0089] FIG. **66** is a cutaway view of the brace of FIG. **57** taken long line **66-66** showing the brace in a locked position.
[0090] FIG. **67** is another elevated rear right side perspective view of the brace of FIG. **42** showing the brace in a deployed configuration for a left-handed user.
[0091] FIG. **68** is an elevated perspective view of a brace for stabilizing a handheld device constructed in accordance with another alternate embodiment of the present invention showing the brace in a deployed configuration for a right-handed user and attached to a handgun.
[0092] FIG. **69** is an elevated rear right side perspective view of the brace of FIG. **68**.
[0093] FIG. **70** is a depressed rear right side perspective view of the brace of FIG. **68**.
[0094] FIG. **71** is an elevated rear left side perspective view of the brace of FIG. **68**.
[0095] FIG. **72** is a depressed rear left side perspective view of the brace of FIG. **68**.
[0096] FIG. **73** is a rear elevational view of the brace of FIG. **68**.
[0097] FIG. **74** is a front elevational view of the brace of FIG. **68**.
[0098] FIG. **75** is a top plan view of the brace of FIG. **68**.
[0099] FIG. **76** is a bottom plan view of the brace of FIG. **68**.
[0100] FIG. **77** is a left side elevational view of the brace of FIG. **68**.
[0101] FIG. **78** is a right side elevational view of the brace of FIG. **68**.

[0102] FIG. **79** is an elevated right side perspective view of the brace of FIG. **68** in a stowed position.

[0103] FIG. **80** is a depressed rear right side perspective view of the brace of FIG. **79**.

[0104] FIG. **81** is an elevated rear left side perspective view of the brace of FIG. **79**.

[0105] FIG. **82** is a depressed rear left side perspective view of the brace of FIG. **79**.

[0106] FIG. **83** is a rear elevational view of the brace of FIG. **79**.

[0107] FIG. **84** is a front elevational view of the brace of FIG. **79**.

[0108] FIG. **85** is a top plan view of the brace of FIG. **79**.

[0109] FIG. **86** is a bottom plan view of the brace of FIG. **79**.

[0110] FIG. **87** is a left side elevational view of the brace of FIG. **79**.

[0111] FIG. **88** is a right side elevational view of the brace of FIG. **79**.

[0112] FIG. **89** is an elevated rear right side perspective view of the brace of FIG. **68** in a deployed configuration for a left-handed user.

[0113] FIG. **90** is a depressed rear right side perspective view of the brace of FIG. **89**.

[0114] FIG. **91** is an elevated rear left side perspective view of the brace of FIG. **89**.

[0115] FIG. **92** is a depressed rear left side perspective view of the brace of FIG. **89**.

[0116] FIG. **93** is a rear elevational view of the brace of FIG. **89**.

[0117] FIG. **94** is a front elevational view of the brace of FIG. **89**.

[0118] FIG. **95** is a top plan view of the brace of FIG. **89**.

[0119] FIG. **96** is a bottom plan view of the brace of FIG. **89**.

[0120] FIG. **97** is a left side elevational view of the brace of FIG. **89**.

[0121] FIG. **98** is a right side elevational view of the brace of FIG. **89**.

[0122] FIG. **99** is an exploded view of the brace of FIG. **79**.

[0123] FIG. **100** is a cutaway view of the brace of FIG. **83** taken along line **100-100** showing engagement of shallow recesses in the distal end of the pivotable arm by protuberances on corresponding opposing surfaces of the slot in the elongated body.

DETAILED DESCRIPTION OF THE INVENTION

[0124] While the making and using of various embodiments of the present invention are discussed in detail below, it should be appreciated that the present invention provides many applicable inventive concepts that are embodied in a wide variety of specific contexts. The specific embodiments discussed herein are merely illustrative of specific ways to make and use the invention and do not delimit the scope of the invention. Those of ordinary skill in the art will recognize numerous equivalents to the specific apparatus and methods described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims.

[0125] To facilitate the understanding of the embodiments described herein, a number of terms are defined below. The terms defined herein have meanings as commonly understood by a person of ordinary skill in the areas relevant to the present invention. Terms such as “a,” “an,” and “the” are not intended to refer to only a singular entity, but rather include the general class of which a specific example may be used for illustration. The terminology herein is used to describe specific embodiments of the invention, but their usage does not delimit the invention, except as set forth in the claims. The term “when” is used to specify orientation for relative positions of components, not as a temporal limitation of the claims or apparatus described and claimed herein unless otherwise specified. The term “lateral” denotes a side to side direction when facing the “front” of an object.

[0126] In the drawings, not all reference numbers are included in each drawing, for the sake of clarity. In addition, positional terms such as “vertical,” “horizontal,” “above,” “below,” “upper,” “lower,” “side,” “top,” “bottom,” and other orientation terms refer to the apparatus when in the orientation shown in the drawing. A person of skill in the art will recognize that the apparatus can assume different orientations when in use.

[0127] The phrase “in one embodiment,” as used herein does not necessarily refer to the same embodiment, although it may. Conditional language used herein, such as, among others, “can,”

“might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

[0128] Turning now to the drawings, wherein like reference numbers refer to like elements, there is illustrated in FIGS. 1 through 10 an embodiment of a brace 10 mountable to a handgun 90 or other handheld device to assist a user in stabilizing the handgun or other handheld device on a user's forearm 92 without straining the user's arm, hand, or wrist. The brace 10 aids a user in stabilizing a handgun or other handheld device in both vertical and horizontal planes with only one hand by releasably engaging two adjacent sides of the user's forearm with a selectively deployable, directionally rigid forearm support member 34 to simultaneously counterbalance the weight of the handgun and reduce lateral rotation of the handgun around the grip. In this way, the brace 10 can counteract the forward tilt exhibited by large and front heavy handguns 90 due to the weight of such weapons being centered at a location forward of their grip, as well as recoil forces generated by a handgun during firing and other commonly encountered environmental forces, such as crosswinds. The brace 10 provides greater and more consistent steadying support than friction alone, without the use of complicated securement straps or uncomfortably tight cuffs that can restrict a user's range of motion. Quickly and easily operable with only one hand, the brace 10 is particularly advantageous for persons who require assistance or otherwise have difficulty with firmly gripping or steadying a handgun for a sufficient period of time to accurately fire the handgun.

[0129] In one embodiment, the brace 10 comprises a body 12, a first arcuate arm 14 extending from the body 12, and a second arcuate arm 16 pivotally connected to an end of the first arm 14 opposite the body 12. The body 12 includes a front end 18, a rear end 20 opposite the front end 18, opposing left and right sides 11, 13 extending between the front end 18 and the rear end 20, and a passage 19 extending axially completely through the body 12 between the front end 18 and the rear end 20. Passage 19 provides for telescopic insertion therein of a portion of a handgun 90 or other handheld device to secure or mount the brace 10 to the handgun or other handheld device. As shown in FIG. 1, a handgun 90 can include a receiver extension 94 or similar support structure that extends rearwardly from a rear end of the handgun 90. The brace 10 is mounted or secured to the receiver extension 94 of the handgun 90 by inserting the receiver extension 94 through the passage 19 from the front end 18 of the body 12. The body 12 is sized to permit the receiver extension 94 to extend completely through the passage 19 and outwardly from the rear end 20 of the body 12. This is desirable in order to enable a user to selectively mount the brace 10 at any location along the length of the receiver extension 94 suited to the length of the user's forearm or personal preference. The body 12 is formed as an upwardly extending lobe including a selectively releasable clamping member 22 and one or more clamp screws 24 operable to retain the receiver extension 94 in the passage 19. Other suitable means of retaining the receiver extension 94 in the passage 19, including but not limited to a set screw, will be apparent to those of ordinary skill in the art and are encompassed within the scope of the invention.

[0130] It is to be understood that the brace 10 is mountable to handguns that do not include a receiver extension 94 such as that depicted in FIG. 1. The brace 10 is mountable to any handgun or other handheld device having a generally cylindrical support structure extending rearwardly from a rear end thereof by telescopically receiving the support structure in the passage 19. It is also contemplated that a suitable support structure comprising a cylindrical or tubular member receivable in the passage 19 can be provided for attachment to a handgun or other handheld device that otherwise lacks a generally cylindrical support structure upon which the brace 10 may be

mounted. For example, a support structure receivable in the passage **19** to mount or secure the brace **10** to a handgun that does not natively include a receiver extension **94** or similar support structure can comprise a spud secured to the rear end of the handgun. In other implementations, a support structure receivable in the passage **19** to mount or secure the brace **10** to a handgun that does not natively include a receiver extension **94** or similar support structure can comprise a spud secured to a mounting bracket that is secured to the rear end of the handgun. Once secured directly or indirectly to the rear end of the handgun, the spud can provide a suitable support structure to which the brace **10** may be attached by receiving the spud within the passage as previously disclosed with respect to receiver extension **94**.

[0131] The first and second arms **14**, **16** of brace **10** each have a proximal end **26**, **30** opposite a distal end **28**, **32**. The first arm **14** is coupled at its proximal end **26** to a side **11** of the body **12**, and extends generally downward from the body **12** such that the distal end **28** of the first arm **14** is positioned below the body **12**. The distal end **28** of the first arm **14** includes a projection **29**, and the proximal end **30** of the second arm **16** includes a recess **31** in which the projection **29** is received. The second arm **16** is pivotally connected at its proximal end **30** to the distal end **28** of the first arm **14** opposite the body **12** by pivot pin **25** to form a knuckle joint such that the second arm **16** is selectively moveable between an open (i.e., deployed) position as best shown in FIG. **2**, and a closed (i.e., non-use or stowed) position as best shown in FIG. **10**. In a particularly embodiment, pivot pin **25** is a roll pin.

[0132] When the second arm **16** is in the open or deployed position, the first and second arms **14**, **16** form a forearm support member **34** having a continuous support surface **35** shaped to receive and engage a user's forearm **92**. The pivotally connected ends of the first and second arms **28**, **30** are shaped such that a portion of the proximal end **30** of the second arm **16** engages a portion of the distal end **28** of the first arm **14** to prevent the second arm **16** from rotating more than about ninety degrees around the pivot pin **25** from the first arm **14**, as shown in FIG. **3**. This helps the forearm support member **34** to remain downwardly rigid and brace the handgun against the directional forces encountered during use of the handgun.

[0133] Specifically, when the user's forearm **92** is received in the forearm support member **34** against the support surface **35**, the portion of the forearm support member **34** formed by the second arm **16** is positioned below the user's forearm **92** so that when the downward moment produced by the forward end of an attached front-heavy handgun **90** biases the rear of the handgun **90** and the forearm support member **34** upward against the user's forearm **92**, the forearm support member **34** remains rigid to counterbalance the weight of the handgun **90** with the user's forearm **92** and limit upward movement of the rear of the handgun. At the same time, the portion of the forearm support member **34** formed by the first arm **14** is positioned laterally to a side of the user's forearm so that the user's forearm can also assist in preventing lateral rotation of the handgun around the grip in at least one direction. Body **12**, which extends laterally over a portion of the user's forearm **92** when the user's forearm is received on the support surface **35**, provides an additional counterbalancing force to reduce muzzle flip when the handgun is fired. Thus, in this way, the brace **10** can assist a user in stabilizing a handgun in multiple directions when the user's forearm **92** is received against the support surface **35** of the forearm support member **34**.

[0134] When the handgun is not in use, the second arm **16** can be moved to the stowed or non-use position to reduce the profile of the brace **10** so that the handgun **90** can be stored more compactly. When the second arm **16** is in the stowed position, the free or distal end **32** of the second arm **16** is received against the side **13** of the body **12** opposite the side **11** from which the first arm **14** extends. The side **13** of the body **12** against which the free or distal end **32** of the second arm **16** is receivable includes two recesses **36**, and the distal end **32** of the second arm **16** includes two prongs **33** shaped to engage recesses **36** to retain second arm **16** in the non-use position. The free or distal end **32** of the second arm **16** also includes a selectively releasable fastening mechanism operable to lock prongs **33** in recesses **36** and thus retain second arm **16** in the stowed position when the

handgun brace **10** is not in use. As more clearly shown in FIGS. **9** and **10**, the fastening mechanism is a push button release including a latch bar **38**, a spring **39**, and a bolt **40**. Each arm **14**, **16** can also include a sling loop **27** defined through a portion of the arm in which a sling is receivable to assist a user with carrying the handgun brace and an attached handgun.

[0135] The body **12** of the brace **10** can also include a limited rotation quick detach sling swivel socket **17** which can receive and engage a quick detach sling swivel, such as a standard push button release sling swivel. As such, the brace **10** can provide a sling swivel attachment point on a handgun **90** enabling a user to attach a sling to the handgun **90** and thus more easily carry the handgun **90** when the sling is attached to the handgun and positioned about the user's person.

[0136] Alternative embodiments are possible and within the scope of the invention. For example, first arm **14** and body **12** can be formed as a single unitary piece, or as separate pieces that are subsequently coupled together. Additionally, the body can be a vertically elongated body formed in part by the first arm, which is fixed in a given position. Furthermore, although the brace **10** is shown in an orientation suitable for a right-handed user, the brace **10** can alternatively be placed in a mirrored configuration such that the brace **10** is suitable for a left-handed user. Alternatively, the brace **10** can be made completely ambidextrous by providing additional recesses **36** in side **11** of the body **12** opposite recesses **36**, prongs **33** or other geometry at the proximal end **26** of first arm **14** that can mate with recesses **36** in side **11**, and a selectively releasable fastening mechanism operable to releasably lock the proximal end **26** of first arm **14** to the side **11** of the body **12** when the brace is not in use or when the second arm **16** of the brace is in a deployed position.

[0137] In other embodiments, the second arm **16** can be pivotally connected to the first arm **14** so that when the second arm **16** is in the stowed position, a rear surface of second arm **16** is received against a front surface of the first arm **14** and the distal end **32** of the second arm **16** is positioned directly below body **12** such that the first and second arms **14**, **16** are vertically parallel.

[0138] In additional embodiments, passage **19** can be a blind passage that does not fully extend through body **12** between the front and rear ends **18**, **20**. Rather, passage **19** can extend through the front end **18** of body **12** and terminate at a position within body **12** proximate to rear end **20**. In such case, receiver extension **94** does not extend beyond the rear end **20** of body **12**.

[0139] In some embodiments, pivot **25** can include a biasing member such as a torsion spring (not shown) which can bias the second arm **16** in either the open or closed position. In some embodiments, the second arm **16** can be biased in the open position, with the torsion spring being placed in either tension or compression as the second arm **16** is moved to the closed position. The engagement of the two prongs **33** and the recesses **36** can resist the biasing force of the torsion spring and retain second arm **16** in the closed position until the user disengages the second arm **16** from body **12**. In other embodiments, pivot **25** can be configured so that friction is produced between the first arm **14** and the second arm **16** such that the second arm **16** only rotates with respect to the first arm **14** when the user applies a force to the second arm **16** to overcome the friction. For example, protrusion **29** of first arm **14** can include a circular recess formed around the aperture in protrusion **29** through which pivot pin **25** extends. A rubber o-ring or the like can be disposed in the circular recess such that compression of the o-ring creates smooth friction between first arm **14**, second arm **16**, and pivot pin **25**. As such, second arm **16** can be retained in any position relative to the first arm **14**, including both the open and closed positions.

[0140] In still other embodiments, brace **10** can include a sear and detent assembly coupled to the first arm **14** and the second arm **16**. The sear and detent assembly can be alternated between a first position where motion of the second arm **16** is prevented, and a second position where motion of the second arm **16** is allowed. In some embodiments, the first arm **14** and the second arm **16** can be configured such that the sear and detent assembly can only be placed in the first position when the second arm **16** is in the open position, such that the sear and detent assembly can selectively retain the second arm **16** in the open position.

[0141] In another alternative embodiment, handgun brace **10** can be mounted to a light weight

handgun such as a GLOCK® brand pistol to dramatically reduce muzzle flip during firing of the handgun. A support structure such as an extension tube can be mounted to a portion of the rear of the handgun, such as the bottom of the grip, so that the extension tube extends rearwardly from the handgun and is receivable in passage **19**. The brace **10** can be mounted to the extension tube in an upside down orientation from the orientation depicted in FIG. **1** such that pivot **25** is located above the body **12**. As such, when second arm **16** is in the deployed position and a user's forearm **92** is received against support surface **35**, the forearm support member **34** is positioned above the user's forearm to provide a counterbalancing force which can prevent the front of the handgun from tilting upward in response to recoil generated by firing the handgun. This can provide increased muzzle control for better target reacquisition and improved successive shot placement as well as reduce the risk of injury and discomfort to physically disabled and novice users resulting from recoil. In such a configuration, body **16** extends laterally under a portion of the user's forearm when the user's forearm is received against the support surface **35**. Thus, when the weight of the handgun biases the body **12** upward, the body **16** engages the user's forearm to help support the weight of the handgun. Therefore, brace **10** can be used with light weight handguns to help support the weight of the handgun on the shooter's arm as well as prevent relative motion of the handgun in multiple directions.

[0142] Turning now to FIGS. **11** through **13**, wherein like reference numerals refer to like elements of previously disclosed embodiments, there is depicted an alternate embodiment of a brace **10b** constructed in accordance with the present invention. Unlike brace **10**, the front end **18** of the body **12** of brace **10b** includes two laterally opposed recesses **49**, each spaced to an opposite side of passage **19**. Brace **10b** further comprises an elongated mounting body **42** having a front end **44**, a rear end **46**, a main passage **43** extending axially completely through the mounting body between the front end **44** and the rear end **46**, two secondary passages **45** that extend axially completely through the mounting body between the front end **44** and the rear end **46**, and two support rods **48**. The secondary passages **45** are each formed on opposite sides of the main passage **43**. Main passage **43** provides for telescopic insertion therein of a handgun support structure such as a receiver extension to secure or mount the mounting body **42** to the handgun as previously disclosed.

[0143] Recesses **49** are shaped to receive and engage support rods **48** to secure or mount the body **12** to the support rods **48**. Secondary passages **45** in mounting body **42** provide for telescopic insertion therein of support rods **48** to attach body **12** to mounting body **42**. Thus, when support rods **48** are received in recesses **49**, body **12** can be attached to mounting body **42** by inserting the support rods **94** through secondary passages **45** from the rear end **46** of the mounting body **42**. Support rods **18** are frictionally retained in secondary passages **45** and are sized to extend completely through the secondary passages **45** beyond the front end **44** of the mounting body **42**. Support rods **48** are axially slidable through secondary passages **45** to permit body **12** to be moved toward or away from the mounting body **42** so a user can adjust the overall length of the brace **10b**.

[0144] Turning now to FIGS. **14** through **41**, wherein like reference numerals refer to like elements of previously disclosed embodiments, there is depicted another alternate embodiment of a brace **10c** constructed in accordance with the present invention. Like brace **10**, brace **10c** comprises a body **12c**, a first arcuate arm **14** extending from the body **12c**, and a second arcuate arm **16** pivotally connected to the first arm **14**. However, body **12c** omits clamping member **22** and clamp screws **24** of brace **10**. Brace **10c** also substitutes elongated body **12c** for body **12** of brace **10**. Brace **10c** additionally substitutes protrusion **50** for recesses **36** on side **13** of brace **10**, as well as passage **19c** for passage **19** of brace **10**. Passage **19c**, unlike passage **19** of brace **10**, includes a channel **37** extending longitudinally along the bottom of the passage within body **12c**. In some embodiments, passage **19c** can extend longitudinally completely through the body **12c** between the front end **18** and the rear end **20**. In other embodiments, passage **19c** can be a blind passage extending within the body **12c** through front end **18** and terminating at a position within the body

12c proximate to but forward of rear end **20**.

[0145] Additionally, the prongs **33** included on the distal end **32** of second arm **16** of brace **10** are omitted from the second arm **16** of brace **10c**. Instead, the distal end **32** of second arm **16** of brace **10c** includes an aperture **41** in which protrusion **50** of body **12c** is receivable, as best shown in FIGS. **27** and **28**. The distal end **32** of second arm **16** of brace **10c** also includes a spring-loaded push button release **52** operable to releasably lock second arm **16** in a stowed position against the side **13** of body **12c** and thus prevent second arm **16** from being moved into a deployed position when protrusion **50** is received in aperture **41** and spring **53** biases the push button release **52** forward to engage the protrusion **50** as shown in FIG. **28**. The second arm **16** can be unlocked and moved into a deployed position upon the application of rearward pressure to the push button release **52** sufficient to compress spring **53** and cause the push button release **52** to disengage the protrusion **50** as shown in FIG. **27**.

[0146] Handgun brace **10c** also comprises a novel support structure **70** designed to permit a user to telescopically mount body **12c** of brace **10c** to a handgun **90**. As shown in FIG. **25**, support structure **70** includes a generally cylindrical body having a rear end **71** opposite a threaded forward end **72** opposite, and a downwardly projecting rib **74** extending along a portion of the length of the body between the rear end and the forward end **72**. Support structure **70** can be attached to any handgun **90** including a native receiver extension **94** in the same way that receiver extension **94** is mountable to handgun **90**, namely, by screwing the threaded forward end **72** of support structure **70** into a threaded aperture located in the rear end of the handgun. Body **12c** is mountable on support structure **70** in the same way that body **10** is mountable on receiver extension **94**, namely, by inserting support structure **70** through passage **19c** from the front end **20** of body **12c**.

[0147] Rib **74** of support structure **70** includes a row of blind cross notches **75** formed in a side of the rib **74**. The intervening portions of the rib **74** between each adjacent cross notch **75** are laterally inset from the exterior surface of the side of the rib **74** to form a lip or lug stop **76** at the forward and rearward end of the row of cross notches **75**. Cross notches **75** are shaped and dimensioned to preclude insertion of the round locking pin included on currently available telescoping buttstocks for AR-style rifles which attach to such rifles via a receiver extension **94**. In this way, the brace **10c** prevents a user from using support structure **70** to improperly mount a rifle buttstock to the handgun **90**, which is prohibited.

[0148] Brace **10c** also comprises a three-stage latch assembly selectively moveable between a locked position, an adjustment position, and a removal position. The latch assembly includes a latch **60** which is received in compartment **63** extending laterally through side **13** of body **12c** at a location forward of first and second arms **14**, **16**, a retaining pin **61**, and a two detent springs **69**. Latch **60** has a thin rectangular base **68**, a button face **62** extending upwardly from one end of the base **68**, two lugs **64** extending upwardly from the other end of the base **68** opposite the button face **62**, and a space **66** between button face **62** and lugs **64**. Each lug **64** includes a bore **65** sized to receive a detent spring **69** therein such that when the latch assembly is assembled into body **12c** as shown in FIGS. **27** through **29**, detent springs **69** received in bore **65** contact an interior wall of body **12c** and bias latch **60** laterally toward side **13**. To prevent detent springs **69** from ejecting latch **60** out of body **12c**, base **68** of the latch **60** includes a retaining pin slot **67** in which retaining pin **61** is received to retain latch **60** inside body **12c**. Lugs **64** are shaped and sized to be received in and engage the cross notches **75** on rib **74** so as to selectively lock body **12c** in one of a plurality of preselected positions along the length of rib **74** when support structure **70** is received in passage **19c**. Space **66** has a width equal to or greater than the width of rib **74** so as to permit rib **74** to slide forward and rearward through the space when the lugs **64** are not engaged with cross notches **75**.

[0149] As shown in FIG. **27**, the latch **60** is in a locked or rest position when the support structure **70** is received in passage **19c** of body **12c** and lugs **64** are received in a pair of cross notches **75** in the rib **74**. The latch **60** is maintained in a locked position by detent springs **69**, which bias lugs **64** laterally to engage cross notches **75** of rib **74** and thus releasably lock body **12c** in a one of a

plurality of preselected positions on support structure **70** so that body **12c** cannot be removed from support structure **70**. When the latch **60** is in the locked position, the button face **62** is flush with the exterior surface of the side **13** of body **12c** and is ergonomically accessible by a user's thumb. [0150] A user can move the latch **60** into the adjustment position to change the overall length of brace **10c** by laterally depressing button **62** until the button is flush with recessed ledge or stop **15** defined in the lower portion of side **13** of body **12c**, as shown in FIG. **28**. This disengages lugs **64** from cross notches **75** and frees rib **74** to slide axially through space **66** between button face **62** and lugs **64**, enabling the user to move the body **12c** longitudinally forward or rearward along the support structure **70** to selectively position the body **12c** at one of a plurality of locations corresponding to a pair of cross notches **75** that suits the user's preference. However, lugs **64** still extend laterally into channel **37** of passage **12c** far enough that lug stops **76** located at either end of the row of cross notches **75** on rib **74** prevent the body **12c** from being accidentally removed from support structure **70** while the latch **60** is in an adjustment position.

[0151] A user can move the latch **60** into the removal position to completely remove the body **12c** from support structure **70** and disassemble brace **10c** for storage by laterally depressing button face **62** beyond recessed stop **15**, as shown in FIG. **29**. When latch **60** is in the removal position, lugs **64** are completely withdrawn from channel **37** and received in internal compartments **79** such that the lugs **64** do not engage cross notches **75** or contact lug stops **76** when the body **12c** is moved forward or rearward on support structure **70**. A user can remove body **12c** from support structure **70** by sliding body **12c** rearward off support structure **70** until support structure **70** is withdrawn from passage **19c**. Body **12c** can only be removed from support structure **70** while latch **60** is in the removal position. The shape, size and lateral movement of button **62** make latch **60** readily accessible and easily operable with a deliberate press with a user's thumb, finger, or prosthesis, while simultaneously reducing the risk of accidental button **62** activation caused by casual interactions with a user's body or the environment. By requiring a deliberate activation, recessed stop **15** prevents unintentional release of body **12c** from the support structure **70**.

[0152] Turning now to FIGS. **42** through **67**, wherein like reference numerals refer to like elements of previously disclosed embodiments, there is depicted another embodiment of a brace **10d** constructed in accordance with the present invention. Brace **10d** comprises vertically elongated body **12d** and an arm **16** pivotally connected to a lower portion of the elongated body **12d**. The arm **16** is selectively movable between a stowed position wherein the arm **16** is aligned with and received inside the body **12d** as shown in FIG. **53**, and a deployed position as shown in FIG. **42** wherein the arm **16** provides a surface **35** against which a user's forearm **92** is removably receivable to stabilize a handheld device such as a firearm **90** when the elongated body **12d** is engaged with the device.

[0153] Elongated body **12d** includes an upper body portion **12**, a lower fixed arm portion **14**, a front end **18** having a shroud **91** extending therefrom, a rear end **20**, a left side **11**, a right side **13**, and a distal end **28d**. A blind passage **19d** and a pair of recesses **49** are formed in the front end **18** of elongated body **12d**. As in previously discussed embodiments, a receiver extension **94** or similar support structure is removably receivable in the passage **19d** to releasably attach the elongated body **12d** to a handgun **90** or other handheld device. Support rods **48** are likewise removably receivable in one or more of the recesses **49** to releasably attach the elongated body **12d** to a handgun or other handheld device. The presence of both the passage **19d** and recesses **49** in the front **18** of elongated body **12d** advantageously increases the connectivity and versatility of the brace **10d**.

[0154] Elongated body **12** also includes a slot **55** formed in each side **11**, **13** thereof. The arm **16** is removably receivable in either slot **55** (depending on whether the brace **10d** is in a right-handed configuration as shown in FIGS. **42-66**, or a left-handed configuration as shown in FIG. **67**) when the arm **16** is in the stowed position. The slots **55** are separated by a thin wall or partition **57** through which an aperture **59** is defined. However, in some embodiments, the slots **55** can be

joined to form one slots extending completely through the elongated body **12d** from the left side **11** to the right side **13**. Each vertical slot **55** includes a forward surface **54** and an opposing rearward surface **56**. The forward and rearward surfaces **54**, **56** can include recesses **58** formed near the upper ends of the slots **55**, as best shown in FIGS. **42** and **43**.

[0155] Pivotal arm **16** includes a distal end **32** and a proximal end **30**. The proximal end of the arm **16** can be pivotally connected to the distal end **28** of elongated body **12d** with a shoulder bolt **25b** or other fastener about which arm **16** can pivot. In some embodiments, the shoulder bolt **25b** can also be used to couple a sling swivel socket **25c** to elongated body **12d**. The proximal end **30** of arm **16** can include a recess **31** and a pair of stops **87** configured to prevent arm **16** from pivoting beyond a predetermined angle away from elongated body **12d** by engaging partition **57**.

[0156] The distal end **32** of arm **16** includes a pair of flexible protrusions **80** which interface with and releasably engage recesses **58** defined in the forward and rearward surfaces **54**, **56** of slot **55** in elongated body **12d** to retain the arm **16** in the stowed position when the arm **16** is received in slot **55**. In this way, the distal end **32** of arm **16** functions as a snap lock that allows a user to quickly and easily lock the arm **16** inside the body **12d** in the stowed position when the arm **16** is not in use. While in the stowed position, the exterior surface of the arm **16** is flush with the exterior surface of the side **11**, **13** of elongated body **12d** in which the arm **16** is received. This reduces both the silhouette of the brace **10d** and the chance that the brace **10d** will become snagged on a user's clothing or the environment.

[0157] In order to unlock the snap lock and release arm **16** from the slot **55**, brace **10d** further comprises elongated snap lock release member **82**. Release member **82** includes a proximal end **84** and a distal end **86**. The proximal end **84** of release member **82** includes a hinge plate **85** that is pivotally connected to the distal end **28d** of elongated body **12d** by shoulder bolt **25b**. Hinge plate **85** is received in a corresponding recess **31** formed in the proximal end **30** of arm **16** when the hinge plate **85** and arm **16** are pivotally connected to the distal end **28d** of elongated body **12d** by shoulder bolt **25b**.

[0158] The distal end **86** of release member **82** includes a protrusion **88** shaped to extend through aperture **59** in partition **57**. As best shown in FIGS. **64-65**, depressing release member **82** while the arm **16** is locked in slot **55** causes protrusion **88** to move through aperture **59** to disengage flexible protrusions **80** from recesses **58** and thus release arm **16** from retention in slot **55** for subsequent rotation into the deployed position. As shown in FIG. **65**, a ledge **89** formed on protrusion **88** acts as a backstop that contacts partition **57** and prevents the protrusion **88** from being inadvertently backed out of the aperture **59**. This prevents the release member **82** from being inadvertently removed from the slot **55** when the arm **16** is in the deployed position. Protrusion **88** and ledge **89** are configured so that the exterior surface of release member **82** remains flush with the exterior surface of the corresponding side **11**, **13** of elongated body **12d** in which release member **82** is received when the release member **82** is not in use.

[0159] A user can remove the distal end **86** of release member **82** from slot **55** by depressing protrusion **88** with a finger or tool while the arm **16** is in the deployed position so that ledge **89** is lowered below the upper edge of aperture **59** in partition **57**. This allows a user to withdraw protrusion **82** through aperture **59** and pivot release member **82** out of slot **55**. A user can completely remove both the release member **82** and arm **16** from slot **55** by removing shoulder bolt **25b** from the lower end **28d** of elongated body **12d**. The release member **82** and arm **16** can then be replaced in slot **55** in either a right-handed or left-handed configuration at the preference of the user.

[0160] Turning now to FIGS. **68** through **100**, wherein like reference numerals refer to like elements of previously disclosed embodiments, there is depicted another alternate embodiment of a brace **10e** constructed in accordance with the present invention. Like brace **10d**, brace **10e** comprises a vertically elongated body **12e** and an arm **16** pivotally connected to a lower portion of the elongated body **12e** by a pivot or other pivotable fastener such as a bolt **25e** and an externally

threaded barrel nut **105**. The arm **16** is selectively movable between a stowed position wherein the arm **16** is vertically aligned with and received inside the body **12e** as shown in FIG. **79**, and a deployed position as shown in FIG. **69** wherein the arm **16** provides a surface **35** against which a user's forearm **92** is removably receivable to stabilize a handheld device such as a firearm **90** when the elongated body **12e** is engaged with the device.

[0161] Unlike brace **10d**, the front end **18** of elongated body **12e** omits any passage or recesses in which a support structure is removably receivable. Instead, the front or forward end **18** of elongated body **12e** includes a forward extension portion in the form of a beam **96** having a distal end **98** configured to be releasably attached to a handheld device **90e** such as a handgun. In one embodiment, the distal end **98** of the beam **96** can be shaped like a portion of a barrel hinge **98** with a hole **99a** in which a releasable fastener such as a bolt **99b** is removably receivable extending therethrough. The barrel hinge **98** can be received in and pivotably connected to complimentary geometry located on a handheld device such as a handgun **90e**, as exemplified in FIG. **68**. Other configurations will be apparent to ordinarily skilled artisans and are within the scope of the invention.

[0162] Brace **10e** is designed to be fully ambidextrous and allow a user to quickly and easily place the arm **16** in a deployed position for either a left-or right-handed user without first disassembling the brace **10e**. To achieve this, brace **10e** omits partition **57** from brace **10d** and includes a single slot **55** that extends completely through the body **12e** from side to side **11**, **13**. This permits arm **16** to pivot through elongated body **12e** from a deployed position for a right-handed user (see, e.g., FIG. **69**) to a deployed position for a left handed user (see, e.g., FIG. **89**). Stops **87** formed on either side of the proximal end **30** of pivotable arm **16** can abut against a cross bar **97** extending between opposing surfaces **54**, **56** of slot **55** near the distal end **28e** of elongated body **12e** to prevent the arm **16** from pivoting beyond a predetermined angle from either side **11**, **13** of elongated body **12e**.

[0163] To facilitate retention of the arm **16** in the stowed position inside the slot **55** of body **12e** when the brace **10e** is not in use (see, e.g., FIG. **79**), the distal end **32** of arm **16** includes a narrow gap **103** between two opposing tabs **102**, **104**. On the forward surface of foremost tab **102** and rearward surface of rearmost tab **104** is formed a shallow recess **101** sized and shaped to receive a complimentary protuberance **108** formed on the corresponding forward and rearward surfaces **54**, **56** of slot **55**. The protuberances **108** engage the shallow recess **101** of each tab **102**, **104** and releasably retain the arm **16** in slot **55** in vertical alignment with the elongated body **12e** when the arm **16** is in the stowed position. Like brace **10d**, the left and right exterior surfaces or edges of the arm **16** do not extend beyond the exterior surfaces of the left and right sides **11**, **13** of elongated body **12e** when the arm is in the stowed position.

[0164] A user can quickly and easily deploy the arm **16** to the right or left (see, e.g., FIGS. **69** and **89**, respectively) by applying pressure to either side of the arm **16** while the arm is received in the slot **55** and shallow recesses **101** are engaged by the protuberances **108**. The narrow gap **103** in the distal end **32** of the arm **16** allows the tabs **102**, **104** to flex toward each other like a living hinge so that the arm **16** can be easily deployed to either the left **11** or right **13** side of the elongated body **12e**, or moved back into the stowed position using only finger pressure.

[0165] In all embodiments disclosed herein, the brace can be made out of a wide variety of strong, durable, rigid materials, including but not limited to, metal, metal alloys, carbon fibers, reinforced polymers, plastics, synthetic polymers, and wood. In some embodiments, the body and first and second arms **14**, **16** are machined from a metal or a metal alloy such as aluminum or steel, respectively. In other embodiments, the body and first and second arms **14**, **16** are injection molded out of a polymeric material such as reinforced polymer. In still yet other embodiments, the body can be made out of a different material or combination of materials than the first and second arms **14**, **16**. In the embodiment depicted in FIGS. **1** through **10**, the brace is machined from billet aluminum. In the embodiments depicted in FIGS. **14** through **100**, the brace is injection molded

from reinforced polymer.

[0166] Although embodiments of the present invention have been described in detail, it will be understood by those skilled in the art that various modifications can be made therein without departing from the spirit and scope of the invention as set forth in the appended claims.

[0167] For example, although aspects of the invention have been described in the context of use as a brace for stabilizing a handgun on or against a user's forearm, it is to be understood that the invention disclosed herein can also be used to stabilize other handheld devices, including without limitation, metal detectors, vacuums, drills and other tools, on or against a user's forearm.

[0168] This written description uses examples to disclose the invention and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

[0169] It will be understood that the particular embodiments described herein are shown by way of illustration and not as limitations of the invention. The principal features of this invention may be employed in various embodiments without departing from the scope of the invention. Those of ordinary skill in the art will recognize numerous equivalents to the specific procedures described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims.

[0170] All of the compositions and/or methods disclosed and claimed herein may be made and/or executed without undue experimentation in light of the present disclosure. While the compositions and methods of this invention have been described in terms of the embodiments included herein, it will be apparent to those of ordinary skill in the art that variations may be applied to the compositions and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit, and scope of the invention. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope, and concept of the invention as defined by the appended claims.

[0171] Thus, although there have been described particular embodiments of the present invention, it is not intended that such references be construed as limitations upon the scope of this invention.

Claims

1. A brace for stabilizing a handheld device against a user's forearm, comprising: an elongated body adapted to detachably engage a portion of the device; and an arm pivotably connected to the body, the arm selectably moveable between a stowed position wherein the arm is substantially aligned with the body, and a deployed position wherein the arm provides a surface against which a user's forearm is removably receivable to stabilize the handheld device when the body is engaged with the handheld device.
2. The brace of claim 1, wherein the elongated body includes a body and a fixed arm.
3. The brace of claim 1, wherein the arm is substantially adjacent the elongated body when the arm is in the stowed position.
4. The brace of claim 1, wherein the arm is vertically parallel with a portion of the elongated body when the arm is in the stowed position.
5. The brace of claim 1, wherein the arm is pivotably connected to a lower end portion of the elongated body.
6. The brace of claim 5, wherein an end portion of the arm distal to the lower end portion of the elongated body is at least partially received inside the elongated body when the arm is in the stowed position.

7. The brace of claim 1, wherein the arm extends outwardly from a side of the elongated body when the arm is in the deployed position.
 8. The brace of claim 1, wherein the arm is selectably moveable at least partially in a side direction away from the elongated body.
 9. The brace of claim 1, wherein the arm is selectably moveable through a plane substantially normal to an axis extending between a forward surface and a rearward surface of the elongated body.
 10. The brace of claim 1, wherein the elongated body includes a slot in which the arm is removably receivable when the arm is in the stowed position.
 11. The brace of claim 10, wherein the slot extends completely through the elongated body from a first side to a second side of the elongated body.
 12. The brace of claim 10, wherein the elongated body is adapted to releasably retain the arm in the stowed position when the arm is received in the slot.
 13. The brace of claim 10, further comprising: a protrusion extending from an interior surface of the slot; and a recess in which the protrusion is removably receivable defined in a portion of the arm to selectably secure the arm in the stowed position when the arm is received in the slot.
 14. The brace of claim 1, wherein: the elongated body includes an elongated extension portion extending from a forward surface of the elongated body; and the extension portion is adapted to detachably engage a portion of the device.
 15. The brace of claim 1, wherein a portion of the elongated body is configured to receive a portion of the handheld device at least partially therethrough.
 16. The brace of claim 1, wherein the elongated body includes a passage in which a portion of the handheld device is telescopically receivable to attach the brace to the handheld device, the passage extending through at least a portion of the elongated body from a forward end of the elongated body toward a rearward end of the elongated body.
 17. The brace of claim 1, wherein the elongated body comprises a selectably releasable clamping member operable to releasably engage a portion of the handheld device to attach the brace to the handheld device.
 18. The brace of claim 1, wherein the handheld device is a firearm.
 19. A firearm, comprising: a rear end portion; and a brace for stabilizing the firearm against a user's forearm, the brace comprising: an elongated body adapted to detachably engage the rear end portion of the firearm, and an arm pivotably connected to the body, the arm selectively moveable between a stowed position wherein the arm is substantially aligned with the elongated body, and a deployed position wherein the arm provides a surface against which a user's forearm is removably receivable to stabilize the firearm when the elongated body is engaged with the rear end portion of the firearm.
 20. A method of stabilizing a firearm, comprising: providing a handgun and a brace for stabilizing the handgun against a user's forearm, the brace comprising: an elongated body adapted to detachably engage a rear end portion of the handgun, and an arm pivotably connected to the elongated body, the arm selectively moveable between a stowed position wherein the arm is substantially aligned with the elongated body, and a deployed position wherein the arm provides a surface against which a user's forearm is removably receivable to stabilize the handgun when the elongated body is engaged with the rear end portion of the firearm; and holding a grip portion of the handgun while the brace is engaged with the rear end portion of the handgun and the user's forearm is received against the arm.
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